Graphics Programming

Perspective Projection

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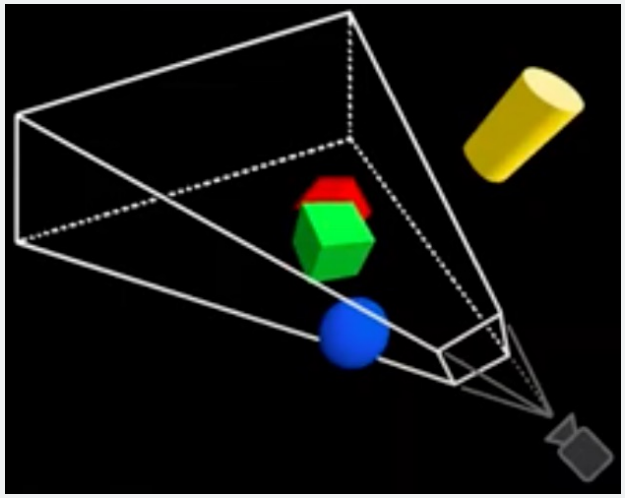
The Visible Frustum

+ Frustum: a truncated pyramid defines visible region
+ Goal: project this volume into clip space
    + Cube: -1< x, y, z < 1



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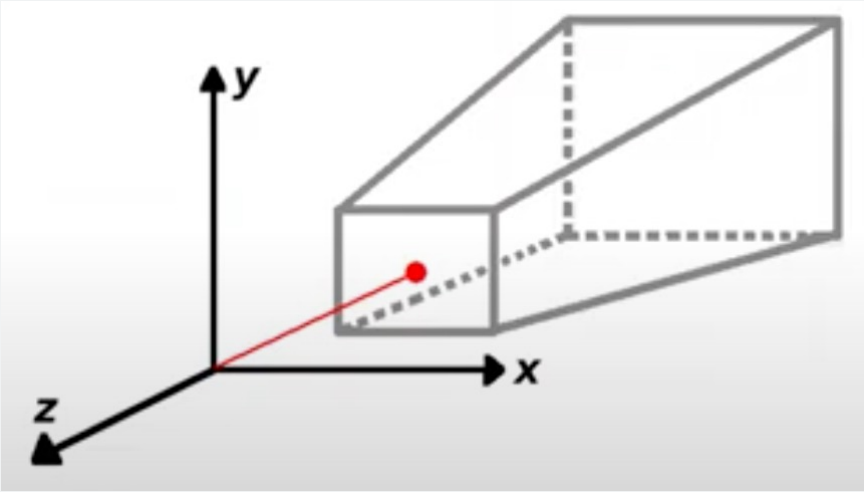
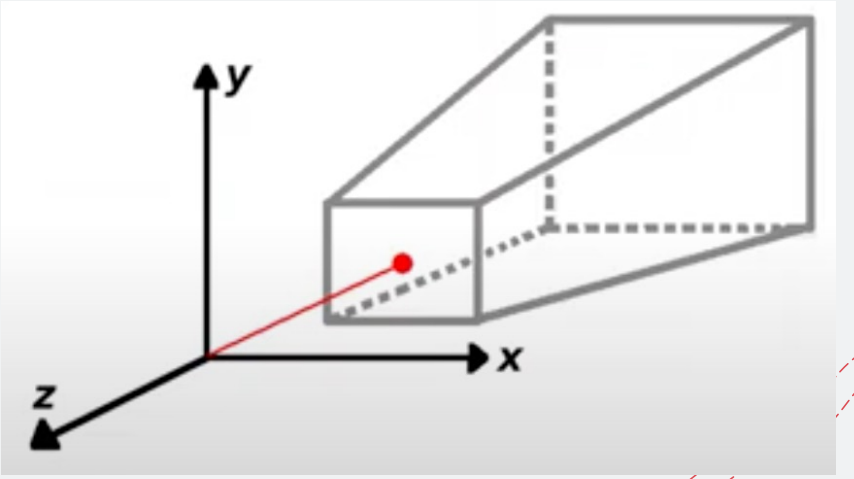
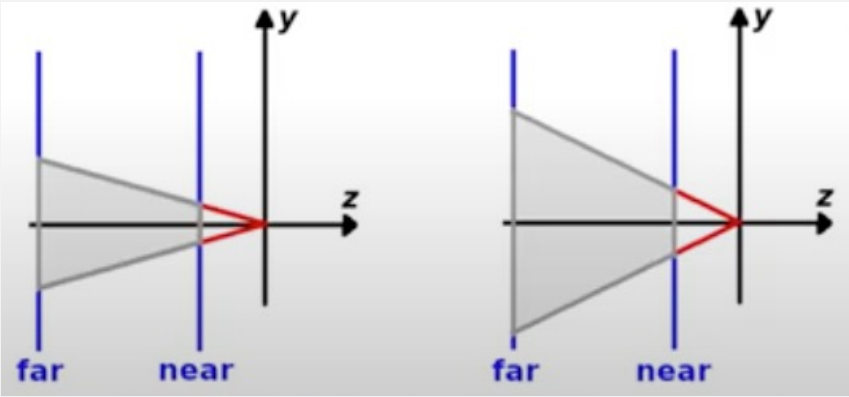


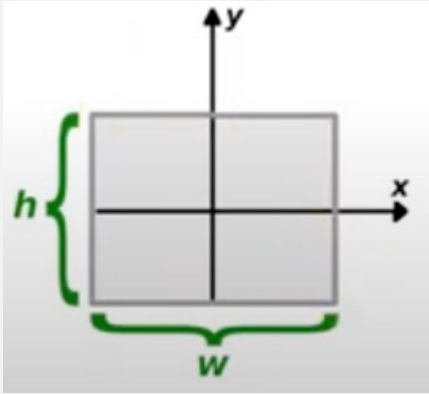
Frustum Parameters

- Near distance
  - The distance from the origin to the near plane
- Far distance
  - The distance from the origin to the farther plane
- Angle of view
  - The angle of the top plane and the bottom plan that is extended to the origin
- Aspect ratio
  - Width_of_rectangle / height_of_ rectangle

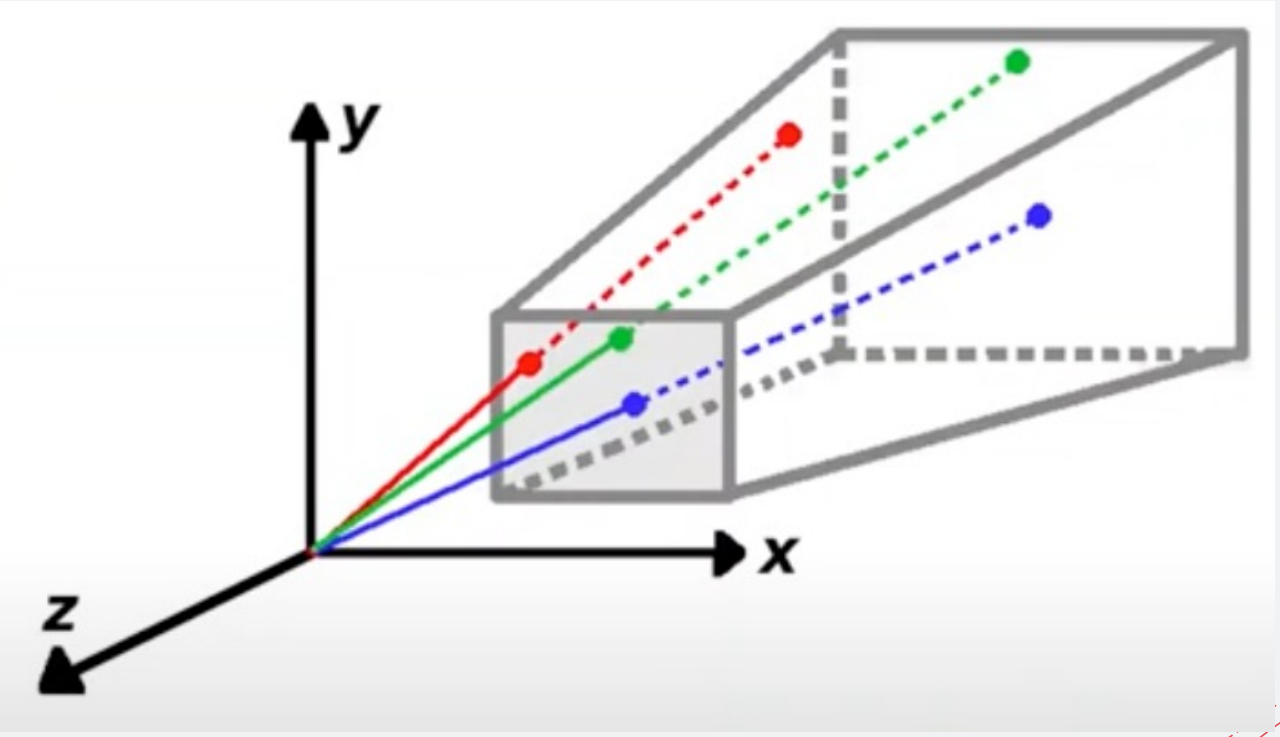


Frustum Parameters



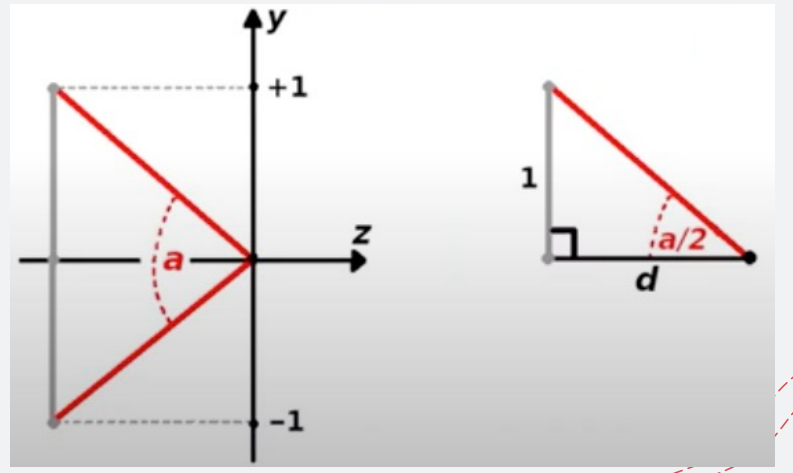


Projecting onto a window



Projection Steps I
-- Adjust projection window

- Adjust projection window to -1 < y 1
- Preserve angle of view
- Distance *d* to window changes
  - $d = 1/\tan(a/2)$



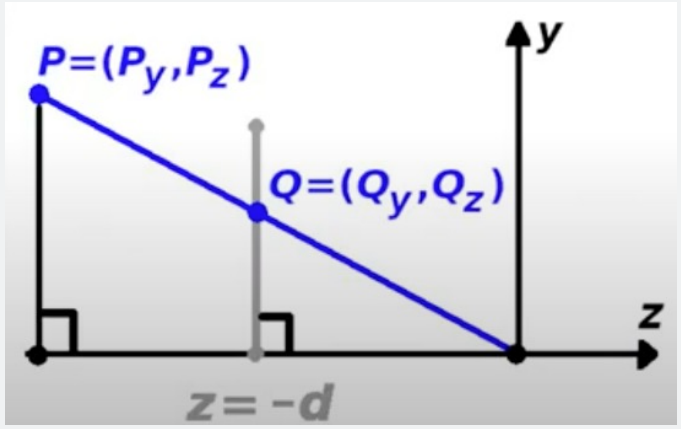
Projection Steps II
-- Formular for projecting y

+Project $P=(P_y, P_z)$ to $Q=(Q_y, Q_z)$

+Similar triangles

  +$P_y/(-P_z) = Q_y/(-Q_z)$

+Since $Q_z=-d$, $Q_Y = d \cdot P_y/(-P_z)$



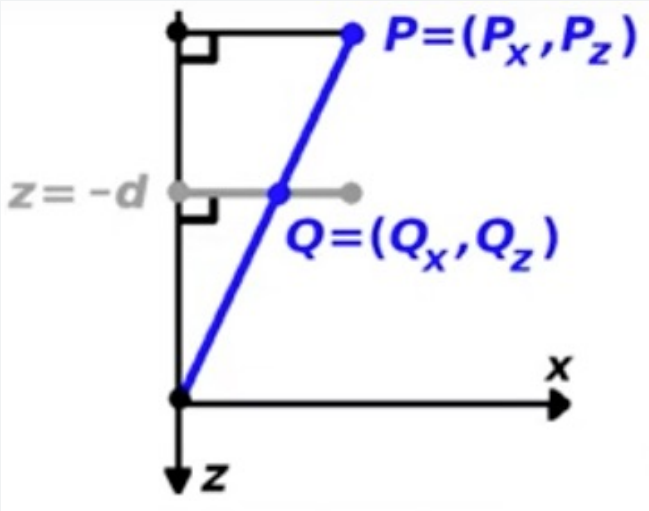
Matrices and Perspective Division

- Division by z-component is not a linear transformation
- Perspective division
  - (x, y, z, w) -> (x/w, y/w, z/w)
- Multiply by the matrix:
  - $$\begin{bmatrix} * & * & * & * \\ 0 & d & 0 & 0 \\ * & * & * & * \\ 0 & 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} P_x \\ P_y \\ P_z \\ 1 \end{bmatrix} = \begin{bmatrix} * \\ d \bullet P_y \\ * \\ -P_z \end{bmatrix}$$
  - Perspective division
    - turns (*, d•P_y, *, -P_z) into (*, d•P_y/(-P_z), *)

Projection Steps III
-- Formular for projecting x

- Project $P=(P_x, P_z)$ to $Q=(Q_x, Q_z)$
- Similar triangles
  - $P_x/(-P_z) = Q_x/(-Q_z)$
- Since $Q_z=-d$, $Q_x = d \cdot P_x/(-P_z)$

- Aspect ration *r*
  - If y-values are in range -1 to 1, x-values include in the render image should range from –*r* to *r*.
  - We need to divide x-values by *r* to scale to range -1 to 1



Update Project Matrix

$$+\begin{bmatrix} d/r & 0 & 0 & 0 \\ 0 & d & 0 & 0 \\ * & * & * & * \\ 0 & 0 & -1 & 0 \end{bmatrix}\begin{bmatrix} P_x \\ P_y \\ P_z \\ 1 \end{bmatrix} = \begin{bmatrix} \left(\frac{d}{r}\right) \bullet P_x \\ d \bullet P_y \\ * \\ -P_z \end{bmatrix}$$

Projection Steps IV
-- Fill out the 3rd row of the projection matrix

- Z-values between near and far distances mapped to range -1 to 1
- X and y components should have no effect on z component
- Write the 3rd row in general terms, then solve for *b* and *c* later

$$\begin{bmatrix} d/r & 0 & 0 & 0 \\ 0 & d & 0 & 0 \\ 0 & 0 & b & c \\ 0 & 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} P_x \\ P_y \\ P_z \\ 1 \end{bmatrix} = \begin{bmatrix} \left(\frac{d}{r}\right) \bullet P_x \\ d \bullet P_y \\ b \bullet P_z + c \\ -P_z \end{bmatrix}$$

- After perspective division
  - Z component became $-b - c/P_z$

A system of equations

- If P_z=-n, then –b –c/ P_z=-1 (P_z=The nearest point)
- If P_z=-f, then –b –c/ P_z= 1 (P_z=The farthest point)

- Equations
  - -b +c/n = 1 and -b -c/f = 1

- Solution
  - c =2•n•f/(n-f)
  - b = (n+f)/(n-f)

Final Project Matrix

$$+\begin{bmatrix} \frac{1}{r \bullet \tan(\frac{a}{2})} & 0 & 0 & 0 \\ 0 & \frac{1}{\tan(\frac{a}{2})} & 0 & 0 \\ 0 & 0 & \frac{n+f}{n-f} & \frac{2 \bullet n \bullet f}{n-f} \\ 0 & 0 & -1 & 0 \end{bmatrix}$$

n: near distance f: far distance a: angle of view r: aspect ratio